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Line printer having user replaceable platen rollers

Abstract

A system and method for user friendly replacement of printer platen rolls includes a roll assembly having a support shaft with a bearing seated in a flange at one end. A crescent notch in a printer chassis receives the shaft into an interior of the notch. The assembly is moved axially so that a flange neck nests in the notch with a flange lip contacting a surface of the chassis. A lock contacts the lip to prevent axial motion of the assembly when installed. An opposed side of the shaft is placed in a bearing on an opposite side of the chassis.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) This application relates generally to a printer feed mechanism for printing a web stock, such as lined or liner-less labels. The application relates more particularly to a media feed system with platen rollers that can easily be replaced by end users without involving a repair technician.

BACKGROUND OF THE INVENTION

(2) In-line printers function to print on a continuous web of print stock, such as label stock. Label printers typically print indicia, such as mailing addresses, onto a label that has adhesive on one side. More recently, liner-less label printing stock is being used.

(3) In-line printing may be accomplished by ink or toner deposition. Direct or indirect thermal printing, as well as impact printing, may also be used. Printing is done on one side for lined adhesive label stock. More recently, printing can be done on both sides using unlined label stock. Dual sided printing uses upper and lower print heads and requires web movement by use of one or more platen roll assembly, such as a top assembly and a bottom assembly.

(4) In-line printers are complex devices with mechanical components that are subject to wear and tear. This is particularly true in high throughput printing operations, such as those used in printing commercial labels. Over time, components may need to be replaced, typically by a qualified technician. This process adds to repair expense, as well as inconvenient downtime until repair is completed.

(5) One component subject to period replacement is a platen roller. A platen roller is a fundamental component found in various types of printers, including dot matrix, impact, thermal, and label

printers. Its primary role is to aid in paper or label handling and ensure accurate paper positioning during the printing process.

(6) The platen roller serves several essential functions. Firstly, it assists in feeding paper or labels through the printer by rotating and advancing them past the printhead or printing mechanism. Secondly, it helps maintain the right amount of tension on the paper or label, preventing paper jams and ensuring consistent printing. Additionally, the platen roller plays a critical role in positioning the paper accurately in relation to the printhead, ensuring precise text and graphics placement on the printed material.

(7) In impact printers like dot matrix printers, the platen roller provides a solid surface against which the printhead strikes, enabling the transfer of ink or impact force onto the paper to create characters and images. In thermal printers, the platen roller may include a heating element. When thermal paper passes over the heated platen, it activates chemicals on the paper, resulting in the production of images and text.

(8) Overall, the platen roller is essential for ensuring the smooth and accurate operation of a printer by facilitating paper or label movement, maintaining tension, and assisting in precise printing. It serves as a critical component in various printing technologies, both impact and non-impact.

(9) Replacing the platen roll often involves disassembling parts of the printer, including removing the printer cover, access panels, and other components. Often times specific tools are required to complete the operation. Technicians know the correct order and method for disassembly and replacement to avoid damage to the printer or injury.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments will become better understood with regard to the following description, appended claims and accompanying drawings wherein:

(2) FIG. 1 is an example embodiment of a print system with user replaceable platen rolls;

(3) FIG. 2 is a perspective view of lower printing assembly 122 of FIG. 1.

(4) FIG. 3 is an expanded view of distal end 212 portion 300 of lower printing assembly 122 of FIGS. 1 and 2;

(5) FIG. 4A is an expanded view of FIG. 3 showing a latch a in a lock position;

(6) FIG. 4B is an expanded view of FIG. 3 showing a latch in an unlocked position;

(7) FIG. 5 is a top perspective view of the upper printing assembly of FIG. 1;

(8) FIG. 6 is a bottom perspective view of the upper printing assembly of FIG. 1

(9) FIG. 7 is a further expanded view of box area 700 of FIG. 6;

(10) FIG. 8A is an expanded view of FIG. 7 showing a latch in an unlock position;

(11) FIG. 8B is an expanded view of FIG. 7 showing the latch of in a lock position;

(12) FIG. 9A is an exploded view of an example embodiment of a platen roll assembly;

(13) FIG. 9B is an exploded view of another example embodiment of a platen roll assembly; and

(14) FIG. 10 illustrates a flowchart of an example embodiment of a method for user friendly replacement of a platen roll assembly.

DETAILED DESCRIPTION OF THE INVENTION

(15) The systems and methods disclosed herein are described in detail by way of examples and with reference to the figures. It will be appreciated that modifications to disclosed and described examples, arrangements, configurations, components, elements, apparatuses, devices methods, systems, etc. can suitably be made and may be desired for a specific application. In this disclosure, any identification of specific techniques, arrangements, etc. are either related to a specific example presented or are merely a general description of such a technique, arrangement, etc. Identifications of specific details or examples are not intended to be, and should not be, construed as mandatory or

limiting unless specifically designated as such.

(16) While example embodiments herein are directed replacement of platen rollers in dual inline printers, it is understood that the structure and methods described herein can be applied to any replaceable roller in any printer type.

(17) In accordance with the subject application, FIG. 1 illustrates an example embodiment of a liner-less print system **100** with user replaceable platen rolls. The liner-less print system **100** includes a pre-feed mechanism **102** comprising a powered drive roller **104** and an idler roller **106**. A reel **112** of liner-less labels **110** is mounted on a rotatable spindle **114**. Liner-less labels **110** from the reel **112** are threaded through the nip **118** between the powered drive roller **104** and the idler roller **106**. The powered drive roller **104** is comprised of rubber or another similar material configured to contact the printable top surface of the liner-less labels **110**. Idler roller **106** advantageously is comprised of silicone or another similar material configured to contact the bottom surface of the liner-less label **110** without adhering to or sticking to the adhesive material on the bottom surface of the liner-less label **110**.

(18) The drive roller **104** and idler roller **106** pass the liner-less labels **110** to the in-line printer **120**. The liner-less labels **110** are pulled into dual sided, in-line printer **120** at a consistent rate, indicia are printed on one or both sides of the liner-less labels **110**, and the label is cut to size by an associated finisher in the in-line printer **120**. In-line printer **120** is comprised of a lower printing assembly **122** for printing on a bottom surface of label stock and upper printing assembly **124** for printing on a bottom surface of label stock. Lower printing assembly **122** is secured to wall **126** so as to form a support chassis with support **128**. Upper printing assembly **124** is secured to wall **126** to form a support chassis with support **132**. Operation of print system **100** is suitably controlled by one or more intelligent controllers, such as intelligent controller **136**.

(19) The liner-less labels **110** can include a printable top surface configured to accept address indicia associated with a shipping label and a bottom surface that includes adhesive. In embodiments, the adhesive is disposed over only a certain portion of the bottom surface, for example around the edges, leaving a second printable area for accepting additional printed indicia such as invoice information for the end customer. In these embodiments, the consumer can remove the label from a received shipment to view the printed indicia on the bottom surface of a liner-less label **110**.

(20) FIG. 2 is a perspective view of lower printing assembly **122** of FIG. 1. Included is lower platen roll assembly **200** that includes support shaft **204** having distal end **208** and a distal end **212**. Distal end **208** is placed into an inner race **216** of bearing **220**. Bearing **220** nests in an interior of flange **222**. Distal end **212** is rotatably connected to bearing **220** in its inner race. Bearing **220** is nested into an interior of flange **222**. Shaft **204** supports radially extending, cylindrical printer platen roll **224**. Flange **222** is pressed into a suitably sized and shaped notch **228**, which can be a crescent notch. While example embodiments herein may refer to a standard bearing seated in a flange neck, it is to be understood that a bearing having a flange integrated into its outer race is also suitably used.

(21) Crescent notch, as used herein, is directed to a notch or opening in a wall or any other barrier that allows something of a smaller diameter to pass through while preventing something of a larger diameter from doing so. In this context, the term may refer to a shape of the opening, which resembles a crescent moon, and its specific function as a selective or restrictive passageway. Latch **232** selectively contacts lip **236** of flange **222** to prevent or allow axial movement of flange **222** relative to notch **228**.

(22) FIG. 3 is an expanded view of distal end **212** showing a platen roll assembly **300** of lower printing assembly **122**. Removal of platen roll assembly from support **128** is accomplished by opening latch **232** by moving it pivotally via its support axis **304**, removing it from contact with lip **236**. Opening the latch enables movement of platen roll assembly **200** in axial direction d.sub.1 removing flange neck **308** from notch **228** while positioning shaft **204** within the notch. A smaller

diameter of shaft **204** allows for passage through an opening of notch **228** by arcuate direction d.sub.2 that includes a radial movement component. Once the assembly is removed, a replacement assembly is added in a complementary fashion. That is, distal end **208** of the replacement assembly is placed in inner race **216** of bearing **220**, the assembly is rotated so that the shaft at distal end **212** passes through notch **228**. Once placed, the assembly is moved axially to nest the flange neck **308** in the notch, and the flange is locked in via rotation and securement of latch **232**. Latch **232** suitably includes a stop **312** that positions the latch in its locking position.

(23) In example embodiments, platen roll assembly **300** may be sold as a replacement unit comprised of the shaft and platen roller, with a flange nesting a bearing affixed to one end. In other example embodiments, the flange or bearing may be reusable.

(24) FIG. **4A** is an expanded view of distal end of platen roll assembly **300** showing latch **232** in a lock position by contacting flange lip **236**. A locked position is coupled with stop **312** in contact with a secondary assembly **316**. FIG. **4B** is an expanded view of distal end of platen roll assembly **300** showing latch **232** in an unlocked position, allowing for removal and replacement of the print roll assembly.

(25) FIG. **5** is a top perspective view of upper printing assembly **124** of FIG. **1**. Included is an upper printing roll assembly **500** that includes an affixed follower gear **504** configured to be actively driven for rotation by one or more drive gear, such as drive gear **508**.

(26) FIG. **6** is a bottom perspective view of upper printing assembly **124** of FIG. **1** showing a view of upper printing roll assembly **500**.

(27) FIG. **7** is a further expanded view of box area **700** of FIG. **6**. Included is upper platen roll assembly **702** that includes support shaft **704** having distal end **708** and distal end **712**. Distal end **712** includes inner race **716** of bearing **720** support shaft **704**, analogously to the description for lower platen roll assembly **400** as detailed above. Additionally, at the distal end **708** a flange/bearing support **724** is configured analogously to that of lower platen roll assembly **400** as detailed above. In the illustrated embodiment, latch **728** is configured to selectively contact a flange lip of the assembly to secure it from axial movement during printer operation, analogously to latch **232** of lower platen roll assembly **400** as detailed above.

(28) FIGS. **8A** and **8B** show latch **728** in an open position and a locked position, respectively.

(29) FIG. **9A** is an exploded view of an example embodiment of a platen roll assembly **900** suitably comprising a platen roll in lower printing assembly **122** or upper printing assembly **124**. Included is platen roll **904** supported by coaxial shaft **908** having distal end **912**. Distal end **912** is configured to receive flange **916** and bearing **914**, nested as shown.

(30) FIG. **9B** is an exploded view of another example embodiment of a platen roll assembly **920** suitably comprising a platen roll in lower printing assembly **122** or upper printing assembly **124**. Included is platen roll **924** supported by coaxial shaft **928** having distal end **932**. Distal end **912** is configured to receive flanged bearing **936** as shown.

(31) FIG. **10** illustrates flowchart **1000** of an example embodiment of a method for user friendly replacement of a platen roll assembly. The process commences at block **1004** and proceeds to block **1008** where a user commences a roller assembly removal operation by locating a platen roll assembly in a printer. Next, at block **1012**, the user rotates a latch handle to disengage the latch from a bearing assembly flange lip. The bearing assembly is then slid axially from a notch, placing an exposed shaft portion to the notch at block **1016**. The exposed shaft is removed from the notch though its gap at block **1020**, and the other end of the shaft is removed from its fixed bearing seat at block **1024**.

(32) A platen roller replacement installation commences at block **1028** where a user place an end of a replacement platen roll shaft into the fixed bearing. The other end of the shaft is passed into the notch via its gap at block **1032**. The bearing assembly is then moved axially to place its flange neck into the notch at block **1036**. The replacement assembly is then latched at block **1040** so that printing with the new platen roll can occur at block **1044**. The process ends at block **1048**.

(33) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the spirit and scope of the inventions.

Claims

1. A replaceable printer platen roll system comprising: a chassis having a first sidewall and a second sidewall, the first and second sidewalls configured to support, in an interior therebetween, opposed, projecting distal ends of a shaft affixed coaxially to a radially extending cylindrical printer platen roll, the chassis configured to receive and support a first outer race of a first bearing on an interior side of the first sidewall, the first bearing having a first inner race configured to axially receive and rotatably support a first distal end of the shaft; a flange assembly including a flange and a second bearing, the flange having a bore coaxially seating a second outer race of the second bearing, the chassis having a notch in the second sidewall configured to axially receive and seat a neck of the flange, the flange further including an inner lip configured to contact the second sidewall when the neck of the flange is seated in the notch, the second bearing having a second inner race configured to receive and rotatably support a second distal end of the shaft; and a latch configured to selectively engage an outer lip of the flange when the flange is seated in the notch such that the flange is axially removable from the second sidewall when the outer lip is disengaged from the latch.
2. The replaceable printer platen roll system of claim 1 wherein the notch includes a gap configured to allow the shaft to pass therethrough.
3. The replaceable printer platen roll system of claim 2 wherein the flange assembly is configured for axial movement corresponding to an applied axial force so as to be selectively placeable in the notch.
4. The replaceable printer platen roll system of claim 3 wherein the flange assembly is configured such that the second inner race of the second bearing receives the second distal end of the shaft when the flange assembly is placed into the gap.
5. The replaceable printer platen roll system of claim 3 wherein the latch is pivotally secured to the second sidewall.
6. The replaceable printer platen roll system of claim 5 wherein the latch includes a radially projecting latch handle.
7. The replaceable printer platen roll system of claim 6 wherein the latch further includes a projection configured to contact a stop when the latch engages the outer lip of the flange.
8. A replaceable printer platen roll system comprising: a chassis having a first sidewall and a second sidewall, the first and second sidewalls configured to support, in an interior therebetween, opposed, projecting distal ends of a shaft affixed coaxially to a radially extending cylindrical printer platen roll, the chassis configured to receive and support an outer race of a first bearing on an interior side of the first sidewall, the first bearing having a first inner race configured to axially receive and rotatably support a first distal end of the shaft; a follower gear affixed coaxially to a second distal end of the shaft; a flange assembly including a flange and a second bearing, the chassis having a notch in the second sidewall configured to axially receive and seat a neck of the flange, the flange assembly further including an inner lip portion configured to contact the second sidewall when the neck of the flange is seated in the notch, the second bearing having a second inner race configured to receive and rotatably support the second distal end of the shaft; and a latch configured to selectively engage an outer lip portion of the flange when the flange is seated in the

notch such that the flange is axially removable from the second sidewall when an outer lip thereof is disengaged from the latch, the follower gear configured to engage a drive gear when the latch is engaged.

9. The replaceable printer platen roll system of claim 8 wherein the notch includes a gap configured to allow the shaft to pass therethrough.

10. The replaceable printer platen roll system of claim 9 wherein the flange assembly is configured for axial movement corresponding to an applied axial force so as to be selectively placeable in the notch.

11. The replaceable printer platen roll system of claim 10 wherein the flange assembly is configured such that the second inner race of the second bearing receives the second distal end of the shaft when the flange assembly is placed into the gap.

12. The replaceable printer platen roll system of claim 10 wherein the latch is pivotally secured to the second sidewall.

13. The replaceable printer platen roll system of claim 12 wherein the latch includes a radially projecting latch handle.

14. The replaceable printer platen roll system of claim 13 wherein the latch further includes a projection configured to contact a stop when the latch engages the outer lip of the flange.

15. A method for changing a platen roll comprising: locating, within a printer, a mounted platen roll assembly, the mounted platen roll assembly including a shaft supporting a coaxially mounted platen roll, the shaft having a first distal end in rotatable contact with a first bearing affixed to a first support, and a flange assembly having a second bearing in a bore thereof, the second bearing having an inner race coaxially secured to a second distal end of the shaft, the flange assembly secured from radial movement by contact between an interior of a crescent notch in a second support and a neck of a flange, and the flange assembly secured from axial movement within the crescent notch by contact between a latch secured to the second support and a lip of the flange assembly; operating the latch to unlatch the flange from the second support; and unmounting the mounted platen roll assembly from the printer by sliding it axially so as to remove the flange assembly from the crescent notch and expose the shaft to the interior of the crescent notch, passing the exposed shaft through an opening of the crescent notch and removing the first distal end of the shaft from contact with the first bearing.

16. The method of claim 15 further comprising: obtaining a replacement platen roll assembly; and mounting the replacement platen roll assembly to the printer by placing a first distal end of a replacement shaft of the replacement platen roll assembly in rotatable contact with the first bearing, passing an exposed portion of the replacement shaft of the replacement platen roll assembly through the opening of the crescent notch, securing the replacement platen roll assembly from radial movement by placing a neck of a flange of a replacement roll assembly in the crescent notch such that the neck thereof contacts the interior of the crescent notch, and securing the replacement platen roll assembly by operating the latch to secure a lip of the flange of the replacement roll assembly to the second support.

17. The method of claim 16 wherein the latch is pivotally mounted to the second support.

18. The method of claim 17 wherein operating the latch includes rotating the latch in a first direction to secure and rotating the latch in a second direction to unlatch.

19. The method of claim 18 further comprising contacting a stop to prevent further rotation of the latch when it is placed a latched position.

20. The method of claim 19 further comprising performing a print operation in conjunction with operation of the replacement platen roll assembly.
